

Message Queues and Sockets

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Definition

Inter-Process Communication (IPC) is a mechanism that allows two or more processes to communicate and exchange data with each other. It helps processes coordinate and share information while running independently.

Message Queues and Sockets

Message Queue is a list of messages stored in the operating system kernel. One process sends messages to the queue and another process reads them in order, so both do not need to run at the same time.

Socket is an endpoint for two-way communication between processes. Sockets allow processes to exchange data on the same computer or across different computers over a network.

Example

Imagine an **online food ordering system** where the customer's app sends an order to the restaurant server using a socket. The server places the order in a message queue, and the kitchen process reads and prepares the orders one by one.

For a **message queue**, the server and kitchen work independently. For a **socket**, the app and server communicate over the network in real time.